ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** is reversed	The field direction reverses.
2	**(a)** to the south	By Fleming's left-hand rule: thumb points to the south.
3	**(d)** increases	Faster change gives a larger emf.
4	**(d)** larger than B	Nearer, stronger.
5	**(d)** perpendicular to the velocity	Force is always perpendicular to velocity.
6	**(d)** the magnet is at rest relative to the coil	No change in the field.
7	**(c)** a north pole	Anticlockwise looks like north.
8	**(d)** zero	No force parallel to the field.
9	**(d)** a larger induced emf	More turns, more emf.
10	**(a)** increases	Proportional to the current.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	At the point of intersection there would be two directions of the field (1); The field has only one direction at a point (1).	1+1
12	Force is zero (1); Force is maximum only when the wire is perpendicular to the field (1).	1+1
13	100 times per second (1); 6000 times per minute (1).	1+1
14	Clockwise appears as a south pole (1); So the left face is the south pole (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Galvanometer deflects when the magnet moves in (1); It deflects in the opposite direction when withdrawn (1); The magnetic field through the coil changes, inducing a current (1).	1+1+1
16	Motor converts electrical energy into mechanical energy; generator does the reverse (1); Motor works on the force on a current-carrying conductor in a field; generator on electromagnetic induction (1); Motor uses a split-ring commutator; an AC generator uses slip rings (1).	1+1+1
17	Straight wire: concentric circles (1); Solenoid: field like a bar magnet (1); Solenoid's field is uniform inside (1).	1+1+1
	OR: (a) Increases with current (1); (b) Increases with turns per unit length (1); (c) Increases with a soft iron core (1).	1+1+1
18	Emf is produced only when the coil cuts the field lines (1); A stationary coil has no change in flux (1); So the current stops when the rotation stops (1).	1+1+1
	OR: Parallel gives each appliance the full voltage (1); The fuse must carry the total current (1); Series fuse cuts off the circuit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Concentric circles around the wire (1); Doubling the current increases the field (1); Doubling the distance weakens the field (1); Right-hand thumb rule: thumb along current (1); Fingers curl in the direction of the field (1).	1+1+1+1+1
	OR: Mains: live, neutral and earth wires (1); Appliances are connected in parallel (1); Geyser current = 2000 / 220 = 9.1 A (1); More than 5 A, so needs the 15 A line (1); Fuse in the live wire melts for excess current (1).	1+1+1+1+1
20	Wind many turns of insulated copper wire over a soft iron rod (1); Connect to a battery (1); Strength increases with current and with number of turns (1); Soft iron core increases it (1); Loses magnetism when current stops, so temporary (1).	1+1+1+1+1
	OR: Short circuit: live and neutral touch (1); Causes: damaged insulation, faulty appliance (1); Overloading: too many appliances on one socket (1); Cause: faulty appliance drawing excess current or accidental rise in voltage (1); Precautions: fuse, earth wire, MCB, good insulation (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Reversing the current	Direction changes.
2	**(c)** Fleming's right-hand rule	Right-hand rule for generators.
3	**(a)** an electromagnet	It becomes magnetised.
4	**(c)** remains the same direction	Two reversals cancel.
5	**(c)** half rotation	Each half turn.
6	**(b)** Soft iron	It demagnetises quickly.
7	**(c)** upward (on the page)	By Fleming's left-hand rule.
8	**(d)** 0.02 s	T = 1 / f.
9	**(b)** it retains magnetism after the current stops	It would become a permanent magnet.
10	**(b)** circular	The force acts as a centripetal force.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	There is no change of the field through the coil (1); No induced current is produced (1).	1+1
12	It melts quickly when excess current flows (1); This breaks the circuit (1).	1+1
13	The field is stronger near the wire (1); Greater field produces more deflection (1).	1+1
14	The coil's sides interchange positions every half rotation (1); The direction of the induced current reverses (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Coil produces a magnetic field like a bar magnet (1); The needle aligns with the field and deflects (1); Reversing the current reverses the poles and the deflection (1).	1+1+1
16	There must be a change in the magnetic field linked with the coil (1); The circuit should be closed (1); Motion of the magnet or coil produces the change (1).	1+1+1
17	Generator: mechanical to electrical (1); Electromagnet: electrical to magnetic (1); Both are useful devices (1).	1+1+1
	OR: The nail becomes an electromagnet (1); It attracts the iron pins (1); Increase the current or the number of turns (1).	1+1+1
18	Red for live, black for neutral, green for earth (1); Potential difference of 220 V (1); The frequency is 50 Hz (1).	1+1+1
	OR: The induced emf increases (1); The rate of change of the field increases (1); The frequency of AC also increases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Force arises from interaction of the wire's field and the external field (1); Forefinger: field (up) (1); Middle finger: current (south) (1); Thumb gives the force (1); Force is directed towards the west (1).	1+1+1+1+1
	OR: DC: one direction (1); AC: periodic reversal (1); Frequency: 50 Hz (1); Period: 0.02 s (1); 100 reversals per second (1).	1+1+1+1+1
20	A rod is suspended between poles of a magnet (1); Current is passed through the rod (1); The rod is displaced (1); Reversing the current reverses the force (1); Reversing the field reverses the force (1).	1+1+1+1+1
	OR: Solenoid carries current in many circular turns (1); Grasp with the right hand, fingers along the current (1); Thumb points towards the north pole (1); One end acts as north and the other as south (1); Field pattern outside is like that of a bar magnet (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.

- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 100 times	Twice per cycle.
2	**(c)** the direction of the field	Definition of a field line.
3	**(b)** the current is increased	Greater current gives a greater force.
4	**(d)** 1/100 s	Frequency is 50 Hz, so reversal occurs 100 times per second.
5	**(c)** the direction of the field	Definition.
6	**(b)** the interaction of two magnetic fields	Wire field and external field.
7	**(a)** earth wire	Earthing.
8	**(b)** field lines are closest	Density of lines.
9	**(b)** tesla	T = N / (A m).
10	**(d)** an air conditioner or geyser	High power.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Live wire is at 220 V relative to the earth (1); Neutral wire is at nearly zero potential (1).	1+1
12	Soft iron gets magnetised and demagnetised quickly (1); Steel remains a permanent magnet (1).	1+1
13	T = 1 / f (1); T = 1 / 50 = 0.02 s (1).	1+1
14	Switching off disconnects the live wire (1); The appliance is then safe to touch (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Conventional current direction is opposite to the electron's motion (1); Apply Fleming's left-hand rule with current to the left (1); Force is downward (on the page) (1).	1+1+1
16	The live wire may touch the body (1); The body gets the high potential (1); Current passes through the person (1).	1+1+1
17	Current drawn = 2000 / 220 = 9.1 A (1); A 5 A socket would be overloaded (1); 15 A circuit is suited for high-power appliances (1).	1+1+1
	OR: Current passes through the coil, the iron core becomes a magnet (1); It attracts the iron armature, the hammer strikes the gong (1); The circuit breaks and the cycle repeats (1).	1+1+1
18	Bring the compass close (1); Pointing end of the needle indicates the opposite pole (1); The N pole of the needle is attracted to the S pole of the magnet (1).	1+1+1
	OR: The field increases (1); Iron gets magnetised by the field (1); Its own magnetism adds to the field (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Moving the magnet towards the coil (1); Moving the magnet away (1); Moving the coil towards or away from the magnet (1); Changing the current in a nearby coil (1); More speed, more turns and a stronger magnet increase the current (1).	1+1+1+1+1
	OR: A turbine rotates the armature coil (1); The coil rotates in the magnetic field (1); The change of the field through the coil induces a current (1); The turbine is driven by steam or water (1); Mechanical energy is converted into electrical energy (1).	1+1+1+1+1
20	Coil rotates between N and S poles (1); First half rotation: current flows in one direction (1); Second half rotation: direction reverses (1); Slip rings and brushes convey the current outside (1); AC output with a frequency of 50 Hz at the power stations in India (1).	1+1+1+1+1
	OR: Oersted: current deflects a compass needle (1); Conclusion: current produces a magnetic field (1); Faraday: moving magnet induces current in a coil (1); Conclusion: changing magnetic field produces current (1); They are opposite effects (1).	1+1+1+1+1

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